

Two barriers to equitable access: WCAG conformance and pre-consent tracking on Ecuadorian and top-ranked university websites

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Abstract

Purpose. Students apply and enrol through university websites, so an obstacle there shuts them out. We studied two such obstacles, whether disabled people can use the site and how it handles personal data, and asked who decides a visitor's protection.

Methods. We examined the home pages of 126 universities: the 63 recognised in Ecuador and 63 top ranked worldwide. For each we read the privacy notice and ran an automated browser check, counting breaches of the Web Content Accessibility Guidelines (WCAG) and every cookie placed before the visitor agreed to anything. We repeated this from Ecuador, Germany, the United Kingdom and the United States.

Results. Ecuadorian universities published less: 61.9% had a privacy notice and 38.1% a privacy contact, against 93.7% and 85.7% among the top ranked. Both groups failed accessibility checks; Ecuadorian sites failed more at the most basic level, 45% against 19%. Tracking before consent was similar in both, yet changed with the visitor's country: 67.8% of sites tracked a visitor in Ecuador and 60.3% one in Germany. The firms supplying the tracking code, not the universities, decided this: Microsoft, TapAd and StackAdapt placed no cookies on European

or British visitors while tracking Ecuadorian and United States ones.

Conclusion. Ecuador’s weaknesses lie in what universities publish and in basic accessibility. But how much tracking a person meets is decided not by the university or its law, but by country lists that advertising firms keep: the same page protects some visitors and not others, purely by location.

Keywords: web accessibility, WCAG conformance, personal data protection, cookie consent, university websites, Ecuador

1 Introduction

The institutional website is the channel through which a prospective student finds programme information, submits an application, completes enrolment and reaches library, tutoring and administrative services. A barrier embedded in that channel is therefore not a cosmetic defect but a restriction on access to the institution. Universal access research has long framed this as a design problem rather than a remediation problem: environments and services should be usable by the widest possible range of people from the outset, and accessibility standards are the operational expression of that principle for digital products [27]. The Web Content Accessibility Guidelines (WCAG), currently in version 2.2 [63], provide the testable criteria, and several legal regimes make them binding, among them the European Accessibility Act [20] and, in the United States, the Americans with Disabilities Act and Section 508 of the Rehabilitation Act [59, 60].

Personal data protection is a second barrier operating on the same channel and, we argue, on the same principle. A visitor who must accept unexplained tracking in order to consult a study programme, or who cannot find out who processes their application data or how to object, faces a cost of participation that is unevenly distributed: it falls hardest on users with least legal literacy, least technical means to refuse and least ability to seek an alternative provider. Read this way, opaque privacy governance and inaccessible interfaces are two mechanisms of the same phenomenon, namely the unequal cost of using a public-facing service. This framing motivates studying them together rather than as two separate compliance checklists.

Empirical work has treated the two literatures separately. On accessibility, audits of university portals report widespread barriers in the highest-ranked universities assessed against WCAG 2.2 [31], in European institutions [29], in the Gulf region [22] and in public online services generally [3, 28], with a long-documented lag in Latin America [1, 8]. On privacy, large-scale measurement has established that written rules and observed behaviour advance at different speeds: the General Data Protection Regulation (GDPR) [19] multiplied notices and banners without proportionate change in tracking [12, 30, 57], consent interfaces frequently fail the legal conditions for valid consent [25, 26, 38, 45], cookie managers often do not block what they claim to block [6, 53], and in higher education specifically, policies have been observed to change to

look compliant rather than to be compliant [40]. To our knowledge, no study has measured both barriers on the same set of institutional sites, with the same protocol and on the same date.

Ecuador is an informative setting for that measurement. Its Organic Law on the Protection of Personal Data (Ley Orgánica de Protección de Datos Personales, LOPDP) entered into force in 2021 and was implemented by regulation only in 2023 [4, 50], placing it in the most recent wave of a regional diffusion process that comparative work has shown to be uneven in commitment and enforcement [9, 10]. The country also has a closed, enumerable population of 63 recognised HEIs, which allows a complete census rather than a sample.

This article addresses four research questions.

- **RQ1.** How do Ecuadorian HEIs and a top-ranked international benchmark group differ in the privacy governance visible on their institutional home page, namely a published privacy notice, a cited legal framework, an enumeration of data-subject rights and an identified privacy contact?
- **RQ2.** How do the two groups differ in observed pre-consent behaviour, that is, in the cookies set before any interaction with a consent interface, when both are observed from the same vantage point outside the European Union?
- **RQ3.** How do the two groups differ in automatically detectable WCAG conformance, and does the difference lie in the volume of failures or in the conformance level at which they occur?
- **RQ4.** Within each group, is measured accessibility associated with observed restraint in pre-consent tracking?
- **RQ5.** Does the pre-consent behaviour of the same page depend on where the *visitor* is located, and if so, does the boundary follow the law that binds the institution or something else?

The contributions are the following. First, a paired measurement of accessibility and pre-consent tracking on 126 university home pages with a single, fully specified and reproducible protocol, released with its configuration, its tracking-cookie taxonomy and its raw output. Second, an inferential reanalysis that separates the differences the design can support from those it cannot, including an explicit equivalence test and a power statement for the null results the study reports. Third, a four-vantage replication of the live audit, from Ecuador, Germany, the United Kingdom and the United States, which converts a legal conjecture into a measurement: the protection a visitor receives is shown to depend on the visitor’s location and to be enforced by advertising vendors rather than by the institutions or by the law that binds them. Fourth, an indicator-by-jurisdiction mapping that separates observed conduct from legal non-compliance, since most of the eleven jurisdictions studied impose no general prior-consent duty for cookies.

The remainder of the article is organised as follows. Section 2 reviews the two literatures and states the gap. Section 3 describes the design, the sampling, the codebook, the audit protocol and the statistical analysis. Section 4 reports the results for each

research question. Section 5 interprets them, derives implications for practice and policy, and states the threats to validity. Section 6 concludes and sets out the future work the present design cannot deliver.

2 Related work

2.1 Web accessibility as a condition of equitable access

Universal design treats accessibility as a property to be built into a service rather than added to it, and media accessibility research has recently examined how that principle interacts with ecological and context-of-use considerations [27]. Applied to higher education, the systematic review by Campoverde-Molina et al [8] covers university website accessibility worldwide and reports persistent non-conformance across regions, methods and time periods. More recent single-study evidence is consistent with that picture: Krittika and Shimray [31] evaluate the top 25 QS-ranked world universities and the top 25 QS-ranked Indian universities against WCAG 2.2 and find failures in both sets; Krejtz et al [29] report on the accessibility information published by European universities; and Fakrudeen [22] reach similar conclusions for the Gulf region. Outside higher education, Aljedaani [3] and Iniesto and Rodrigo [28] document the same dominant error types on public-sector portals, principally insufficient contrast and links without an accessible name. In Latin America, Acosta-Vargas et al [1] and Acosta-Vargas et al [2] provide comparative evidence of a regional lag, and Maldonado-Garcés et al [37] extend the argument from the digital to the physical campus in Ecuador.

Two methodological cautions follow from this literature and shape the present design. First, automated tools demonstrate non-conformance but cannot certify conformance: Fischer et al [23] quantify the coverage of automated checkers and show that they address only a minority of the WCAG success criteria, so an automated pass is not evidence of an accessible site. Second, results depend on the tool, on the rule set enabled and on the rendering conditions, which is why this study reports its full configuration rather than only its tool name.

2.2 Privacy governance and web privacy measurement

The measurement literature on web privacy provides both the instruments and the cautionary findings for the present study. Libert [35] and Englehardt and Narayanan [15] established large-scale browser-based measurement of third-party tracking and showed that the phenomenon is concentrated, persistent and largely invisible to users. After the GDPR became applicable, Degeling et al [12] and Trevisan et al [57] found that notices and banners proliferated without a proportionate reduction in tracking, and Kretschmer et al [30] reached comparable conclusions over a longer window. A second strand shows that the interfaces themselves are frequently non-compliant: Matte et al [38] found that a large share of banners built on the IAB Europe Transparency and Consent Framework registered a consent signal inconsistent with the user's choice; Nouwens et al [45], Gray et al [25] and Habib et al [26] document design patterns that make acceptance the path of least resistance; and Sanchez-Rola et al [53] and Bollinger et al [6] show that consent-management platforms often fail to block the trackers they

purport to gate. On the textual side, privacy policies remain long and hard to read [5, 21].

A third strand, directly relevant here, treats the location of the observer as an experimental variable rather than as a fixed property of the study. Dabrowski et al [11] crawled the same sites from clients in the European Union and in the United States and found that 26.6% of the cookie-setting sites placed persistent cookies for the United States client and none for the European one, an asymmetry that was almost absent in the opposite direction. van Eijk et al [14] crawled from eighteen vantage points and reached a more conservative conclusion, that the top-level domain of the site explains more of the variation in cookie notices than the location of the visitor does, with .com sites the exception. Rasaii et al [52] confirm the pattern on the notice side, reporting consent banners 56% more prevalent when a site is visited from within the European Union. The three studies agree that where a measurement is taken from changes what is measured; they do not agree on what governs the difference.

In higher education, Liu and Khalil [36] review privacy issues in learning analytics and Mutimukwe et al [41] examine online proctoring, both concerned with internal systems rather than the public portal. Closest to the present work, Munot et al [40] analyse how universities reshaped their privacy policies after the GDPR and distinguish compliance achieved by design from compliance achieved by appearance.

2.3 The gap addressed here

Three gaps follow from the two literatures reviewed above, and the design of this study is a response to each of them.

The first is a gap of scope. The two literatures share a subject, the institutional website, and an underlying concern, the cost that a service imposes on the people who must use it, but they have not been brought together on a common set of sites. Accessibility studies do not measure tracking; privacy measurement studies do not audit conformance. Because the two are always measured apart, a question that matters for practice cannot be answered at all: whether an institution that commits to one form of access also commits to the other, or whether the two are independent properties that a single institutional commitment does not deliver. Section 4.5 tests that association directly, which requires both barriers to be measured on the same sites, on the same day, with one protocol.

The second is a gap of vantage point. The geo-differential literature is built almost entirely on an axis between the European Union and the United States, and it attributes what it finds to the site: a site either discriminates by visitor location or it does not. Two consequences follow. Latin America appears in none of these designs, so what a visitor located under a young data-protection regime actually receives has not been measured; and because the unit of attribution is the site, the mechanism is left unresolved, since an asymmetry observed at the site cannot distinguish a decision taken by the operator of the site from one taken by the third parties whose code the site embeds. This distinction is not academic: it determines whether a remedy addressed to universities can work at all. Section 4.6 measures from four vantage points, one of them Ecuadorian, and attributes the difference to named third-party domains rather than to the sites that host them.

The third is a gap of population. Studies of university websites, in both literatures, rest overwhelmingly on ranked or convenience samples, which by construction describe institutions selected for visibility. A complete national population supports a different claim, because it carries no sampling error and no selection on prominence, and because it is the population that a national regulator is actually obliged to supervise. Setting such a population against an elite benchmark under an identical protocol is what allows the differences reported in Section 4 to be read as differences between the two groups rather than as differences between two studies. That combination, two barriers, four vantage points and a complete national population against a benchmark, is what this article contributes.

3 Methods

3.1 Design and unit of analysis

The study is a cross-sectional, two-group comparison carried out in August 2026. The unit of analysis is the *institutional home page and the first-level privacy notice linked from it*. This is a deliberately narrow unit and it is stated explicitly because it determines what the results can mean: a document that exists on a faculty subdomain but is not reachable from the home page is coded as absent, since a prospective applicant arriving at the institutional portal would not find it. Where a first-level notice referred the reader onward to a more detailed statement, the referenced statement was read and coded, but only when it was reachable in one click from the first-level notice.

The comparison is a benchmarking exercise against an aspirational ceiling, not an estimate of a country effect. The benchmark group is not a sample of “the rest of the world”: it is the extreme upper tail of an international ranking distribution, and it differs from the Ecuadorian population in budget, technical staffing and regulatory maturity as well as in country. Every difference reported in Section 4 is therefore open to interpretation as an effect of institutional resources rather than of jurisdiction, and no causal attribution to country is made anywhere in this article.

3.2 Sampling

The Ecuadorian group is a complete census of the 63 universities and polytechnic institutions recognised by the national higher-education authorities, the Secretariat of Higher Education, Science, Technology and Innovation (Secretaría de Educación Superior, Ciencia, Tecnología e Innovación, SENESCYT) and the Council of Higher Education (Consejo de Educación Superior, CES), as listed on 11 August 2026. Institutions created in 2024 and 2025 are included, and are identified as such in Appendix D so that readers can exclude them if they consider the comparison unfair to institutions that have not yet completed their first academic cycle.

The benchmark group was constructed by consensus of three rankings in their most recent editions: QS World University Rankings 2026, Times Higher Education World University Rankings 2026 and the Academic Ranking of World Universities 2025 [51, 54, 56]. From each ranking the top 75 was taken; each institution received the mean of its three positions, with absence from a ranking’s top 75 imputed as position

76; the number of rankings in which an institution appeared took precedence over that mean, so the 46 institutions present in all three head the list and the remaining 17 were drawn, by mean position, from those present in two. The cut-off institution was the University of Queensland and the highest-placed institution excluded was the London School of Economics. The group size was set at 63 to match the Ecuadorian census. Figure E1 in Appendix E shows the resulting distribution by country; the United States contributes 24 institutions, the United Kingdom 7, China 6 and Australia 5, with the remainder spread over eleven further jurisdictions.

The aggregation rule involves three arbitrary choices, namely the imputation constant, the depth of the ranking cut and the lexicographic priority given to ranking coverage. These were not varied, and the sensitivity of the composition to them is reported as a limitation in Section 5.5 rather than claimed as robust.

3.3 Dimensions, operational definitions and codebook

Five documentary dimensions and three live dimensions were coded. The operational definitions are given in full in Appendix A; the essential points are the following.

Privacy notice. Coded as present when a document whose stated purpose is to describe the processing of personal data is reachable from the home page. Coded **P** (partial) when the document exists but its stated scope covers only part of the institution’s processing, for example admissions only, postgraduate programmes only or a mobile application only. Coded **NV** (not verifiable) when the site could not be retrieved or the document was not machine-readable.

Applicable legal framework cited. Coded as present when the notice names a specific legal instrument. Generic formulations such as “applicable law” or “data protection legislation” were coded as absent. In Ecuador the relevant instrument is the LOPDP and its implementing regulation [4, 50]; elsewhere the framework applicable to the institution was accepted, as set out in the indicator-by-jurisdiction mapping of Appendix B.

Data-subject rights listed. Coded as present when the notice enumerates at least three of the rights of access, rectification, erasure, restriction, objection and portability, together with a route to exercise them. A generic mention of “your rights” without enumeration was coded as absent.

Privacy contact or data protection officer. Coded as present when the notice gives a named role or a dedicated address or form for privacy matters. A general institutional contact form was coded as absent. We distinguish throughout between the *controller*, which is the institution itself and is therefore never absent, and the *data protection officer or privacy contact*, which is the designated route through which a data subject can act.

Transport security. The presence of a valid TLS certificate was recorded. This indicator is reported under the heading of transport security and not of data protection: it measures confidentiality in transit and says nothing about what the site does with the data once received. It was verified separately from the automated audit, since the audit browser was configured to ignore certificate errors in order to reach sites with faulty chains.

Live dimensions. Pre-consent cookies, tracking cookies, consent banner and consent-management platform (CMP) were recorded by the automated audit described in Section 3.5, as was WCAG conformance.

3.4 Documentary coding and its verification

The documentary dimensions were coded by the authors from the rendered documents. The study does not report an inter-coder reliability coefficient, because double coding was not performed; this is a material limitation and is stated as such in Section 5.5.

To obtain at least a bounded estimate of coding error, a targeted verification sub-study was carried out on 14 August 2026. All six institutions that an earlier draft of this article named individually were re-examined manually, cell by cell, against the live site. Two discrepancies were found in the 24 cells checked. First, one institution had been coded as publishing no privacy notice when a notice is in fact linked from the home-page footer; the cell was corrected to present, which raises the benchmark group’s privacy-notice count from 58 to 59. Second, one institution had been coded as citing an applicable framework when its notice names no instrument and uses only generic formulations; the cell was corrected to absent, which lowers the benchmark group’s framework count from 48 to 47. Both corrections are incorporated in every figure and table of this article. The sub-study also revealed a systematic effect of the depth-one unit of analysis: at two further institutions, rights and framework are stated one click below the first-level notice and are therefore recorded as absent under our definition even though the content exists. This is a property of the measure, not an error, but it means that the governance indicators reported here should be read as “visible at the first level” rather than as “published anywhere by the institution”.

Two errors in 24 cells is not a reliability estimate, and it is reported only to indicate that the error rate is not negligible. Full double coding with a Cohen’s κ per indicator is the first item of the future work set out in Section 6.

3.5 Automated audit protocol

Each of the 126 sites was visited once, in August 2026, from a residential connection in Ecuador, with a read-only script. The full configuration is given here so that the measurement can be repeated.

- *Browser and driver.* Chromium driven by Playwright [39], launched headless with the flags `--no-sandbox` and `--disable-dev-shm-usage`. A fresh browser context was created per site, so no state carried over between sites.
- *Viewport and user agent.* Playwright’s default desktop viewport of 1280×720 CSS pixels was used, with the user-agent string of Chrome 128 on Windows 10 (64-bit). The viewport is reported because it determines the outcome of the contrast, reflow and target-size criteria.
- *Navigation.* `page.goto` with `waitUntil: 'domcontentloaded'`, a 45 s timeout and one retry on failure, followed by a fixed 4 s wait to allow deferred scripts and consent platforms to load. The context was created with `ignoreHTTPSErrors: true` so that sites with certificate problems could still be audited. No scrolling and no interaction with any banner were performed at any point.

- *Cookie state.* The full cookie jar of the context was read before any interaction. A cookie was classified as *tracking* when its name matched a declared allow-list of analytics, advertising and session-replay families, reproduced in full in Appendix A. The list covers Google Analytics and Google Ads, Meta, TikTok, Microsoft Clarity and Bing, Hotjar, Matomo, Siteimprove, Pinterest, Adobe, Tealium, Baidu, Quantcast, LinkedIn, X and the YouTube and DoubleClick families. Cookies not matching the list, including first-party session and language cookies, were counted in the total but not as tracking.
- *Consent platform and banner.* A CMP was recorded when a signature of one of thirteen named platforms matched the `src` of a script, iframe or link element, when one of a fixed white-list of CMP global objects was defined, or when `window.__tcfapi` was a function. A banner was recorded when a fixed or sticky element of non-zero height below 600 px contained text matching cookie, consent or privacy. Both are heuristics and their false-negative rate was not measured.
- *Accessibility.* axe-core 4.13 [13] was injected into the loaded page and run over the whole document with `resultTypes` set to violations, passes and incomplete. The rule set was selected by tag: `wcag2a`, `wcag2aa`, `wcag21a`, `wcag21aa`, `wcag22a` and `wcag22aa`, plus `wcag2aaa` and `wcag21aaa`. The reference standard is therefore WCAG 2.2 at levels A and AA [63] together with the level-AAA rules of WCAG 2.0 and 2.1 [62]; no WCAG 2.2 level-AAA rules were enabled, because axe-core 4.13 does not implement them.
- *Derived measures.* Each violation was assigned its success criterion, its principle and its conformance level from the axe tags, with the highest level taken when a rule carries several. A site was recorded as *reaching level A* when it produced no level-A violation, and as having *no automatic failure* when it produced no violation at any enabled level. The number of checks axe flags for manual review was recorded per site but is not analysed here.

Two properties of this protocol matter for interpretation and are not incidental. First, enabling the level-AAA rules is a substantive choice: the dominant AAA rule is enhanced contrast (success criterion 1.4.6, requiring a 7:1 ratio), and any statement about the AAA share of failures is conditional on that rule being active. Section 4.4 therefore reports the level distribution both with and without the AAA rules. Second, the single vantage point in Ecuador is what the pre-consent measurement means: what is observed is what an Ecuadorian visitor experiences, not what a visitor located in the European Union would experience.

3.6 Multi-vantage replication

Because the audit described above observed from one location only, it could not distinguish two possibilities with opposite legal readings: that sites treat every visitor alike, or that they present different interfaces depending on where the visitor is. To separate them, the live audit was replicated from four vantage points on 15 August 2026, with every other parameter unchanged.

The Ecuadorian vantage point was a residential connection in Cuenca (ETAPA EP, AS27668) with no VPN. The other three were reached through a commercial

VPN: Frankfurt, Germany (DigitalOcean AS14061 for two passes, Limestone Networks AS46475 for one); London, United Kingdom (Zenlayer AS21859); and Miami, United States (Limestone Networks AS46475). Three passes were run per vantage point, twelve in total, and each site’s outcome was consolidated across passes by majority rule. The public IP address and its geolocation were verified at the start of every pass, after every 25 sites and at the end, giving seven checks per pass; all 84 checks matched the declared vantage point, and the recordings are released with the data. One further pass was aborted automatically by that mechanism when the VPN had not yet switched country, and was discarded.

Two design properties of this replication matter. First, the Ecuadorian and United States vantage points are both outside the European Economic Area but differ in network type, residential against data centre, whereas the German, British and United States endpoints are all data-centre addresses; the German-versus-United States contrast is therefore free of any residential-versus-data-centre artefact. Second, three quantities that cannot depend on consent logic were used as controls: the number of WCAG violations, the number of failing nodes and the volume of cookies that are not attributable to any advertising vendor.

The tracking-cookie taxonomy was extended for this analysis after inspection of the cookie names observed. The original list covered Google, Meta, TikTok, Microsoft Clarity, Hotjar, Matomo, Baidu and several smaller families; it did not cover the LinkedIn Insight Tag, the full Microsoft advertising family, TapAd, StackAdapt, Snapchat or Sourcebuster. The extended list, given in full in Appendix A, is applied by reclassifying the stored cookie names, so no re-measurement was required and every reported figure derives from the same recorded observations.

3.7 Statistical analysis

Proportions are reported with 95% Wilson score intervals. Group differences are reported as risk differences with 95% Newcombe intervals and as odds ratios with 95% Woolf intervals, using the Haldane–Anscombe correction of 0.5 where a cell is empty. Significance is assessed with the two-sided Fisher exact test, which is appropriate throughout given the expected cell counts in the regional breakdown. Because twelve indicators are compared, p values are adjusted by the Holm–Bonferroni procedure and both raw and adjusted values are reported; the adjusted value governs every claim of difference made in this article.

For the indicator on which no difference was found, an equivalence test was performed rather than an argument from a non-significant p value. Two one-sided tests (TOST) were run at margins of 10, 15 and 20 percentage points. The margins were not pre-specified, which is stated openly, and the result is reported as a bound on what the design can establish rather than as a demonstration of equivalence.

Vantage points are compared with the exact McNemar test, which is the appropriate test when the same site is observed under two conditions, and with Cochran’s Q for the four-way comparison; discordant-pair counts and exact p values are reported alongside every such comparison. The association between accessibility and pre-consent tracking is analysed within each group separately as well as pooled, because pooling

two populations with different marginal distributions on both variables invites Simpson’s paradox. Percentages of failing nodes by conformance level are rounded to the nearest integer, which is why the benchmark distribution sums to 101 rather than 100. All analyses were carried out in Python 3 with SciPy; the analysis script is released with the data.

3.8 Ethical considerations

The study involves no human participants and no personal data: the observations are properties of public web pages. It nonetheless carries reputational risk for the institutions observed, and three measures were adopted in response. First, no institution is named in the body of the article as failing an indicator; institution-level results appear only in the appendices, where the coding date, the definition and the limitations of the measure are stated alongside them. Second, all cells for institutions that an earlier draft named individually were manually re-verified, and the two corrections found are reported in Section 3.4. Third, the results are framed as a description of observed conduct on one date, not as a determination of legal non-compliance, since, as Appendix B sets out, most of the jurisdictions studied impose no general duty of prior consent for cookies.

The script observed a conservative load profile, one request sequence per site with no crawling beyond the home page, and did not attempt to circumvent access controls.

4 Results

Table 1 reports every indicator for both groups with the risk difference, the odds ratio and the Holm-adjusted p value; the Wilson intervals for each proportion are given in the text and displayed in Figure 1. Figure 1 displays the same comparison graphically, grouped by domain. Appendices C and D give the institution-level matrices.

4.1 Privacy governance (RQ1)

The governance differences were the largest and the most robust in the study. A privacy notice was reachable from the home page at 59 of the 63 benchmark sites (93.7%) and at 39 of the 63 Ecuadorian sites (61.9%), a difference of 31.7 percentage points (odds ratio 9.08, adjusted $p < 0.001$). The requirement that the notice name a specific legal instrument reduced both figures, to 74.6% and 49.2% respectively (odds ratio 3.03, adjusted $p = 0.045$).

The widest gap was in the route by which a data subject can act. A privacy contact or data protection officer was identified at 85.7% of benchmark sites and at 38.1% of Ecuadorian sites, a difference of 47.6 percentage points (odds ratio 9.75, adjusted $p < 0.001$). The enumeration of data-subject rights differed in the same direction, 74.6% against 55.6%, but this difference did not survive correction for multiple comparisons (adjusted $p = 0.235$) and is reported here as suggestive only.

The regional breakdown of the benchmark group is given in Table 2. It should be read as descriptive: with subgroups of three to five institutions the Wilson interval around a proportion spans more than 50 percentage points, so no regional ranking can

Table 1 Indicator-by-indicator comparison of the benchmark group and the Ecuadorian census ($n = 63$ per group, August 2026). RD: risk difference in percentage points, benchmark minus Ecuador, with 95% Newcombe interval. OR: odds ratio with 95% Woolf interval. p : two-sided Fisher exact test; p_H : Holm-adjusted over the twelve comparisons.

Indicator	Benchmark	Ecuador	RD [95% CI]	OR [95% CI]	p_H
<i>Privacy governance (documentary)</i>					
Privacy notice published	59/63 (93.7)	39/63 (61.9)	+31.7 [+17.6, +44.7]	9.1 [2.9, 28.2]	<0.001
Framework cited	47/63 (74.6)	31/63 (49.2)	+25.4 [+8.4, +40.4]	3.0 [1.4, 6.4]	0.045
Rights listed	47/63 (74.6)	35/63 (55.6)	+19.0 [+2.4, +34.3]	2.4 [1.1, 5.0]	0.235
Privacy contact or DPO	54/63 (85.7)	24/63 (38.1)	+47.6 [+31.3, +60.4]	9.8 [4.1, 23.3]	<0.001
<i>Pre-consent behaviour (live audit)</i>					
Tracking cookies set	43/63 (68.3)	40/63 (63.5)	+4.8 [−11.6, +20.7]	1.2 [0.6, 2.6]	1.000
At least one cookie set	56/63 (88.9)	55/63 (87.3)	+1.6 [−10.2, +13.4]	1.2 [0.4, 3.4]	1.000
Consent banner shown	22/63 (34.9)	15/63 (23.8)	+11.1 [−4.8, +26.3]	1.7 [0.8, 3.7]	0.961
CMP detected	12/63 (19.0)	14/63 (22.2)	−3.2 [−17.2, +11.0]	0.8 [0.4, 2.0]	1.000
<i>Accessibility</i>					
Accessibility statement	47/63 (74.6)	7/63 (11.1)	+63.5 [+47.9, +74.2]	23.5 [8.9, 61.9]	<0.001
No level-A failure	25/63 (39.7)	5/63 (7.9)	+31.7 [+17.2, +44.9]	7.6 [2.7, 21.7]	<0.001
No failure at any level	6/63 (9.5)	0/63 (0.0)	+9.5 [+1.8, +19.3]	14.4 [0.8, 260]	0.193
<i>Transport security</i>					
Valid TLS certificate	63/63 (100.0)	58/63 (92.1)	+7.9 [+0.6, +17.3]	11.9 [0.7, 221]	0.288

Note: percentages in parentheses. “No level-A failure” denotes the absence of automatically detectable level-A violations and is not a claim of WCAG conformance, which automated tools cannot establish [23]. Two cells of the benchmark group were corrected after manual re-verification, as described in Section 3.4.

be supported by these data. Two patterns are nonetheless worth recording because they are large relative to that uncertainty. The Chinese institutions in the benchmark group are the least likely to publish a first-level notice, two of six, and none published an accessibility statement. Continental Europe cites a legal framework at nine of ten institutions, but the ten institutions are not homogeneous in the applicable instrument: two are Swiss and are subject to the revised Federal Act on Data Protection [55] rather than to the GDPR. Similarly, the three Canadian institutions are public bodies governed by provincial public-sector statutes [32, 33, 42], not by the federal PIPEDA [47], which applies to commercial activity; the earlier practice of describing a single Canadian regime is corrected here.

4.2 Pre-consent behaviour (RQ2)

Figure 2 reports the live measurements. Observed from Ecuador, the two groups behaved similarly. At least one cookie was set before any consent interaction at 88.9%

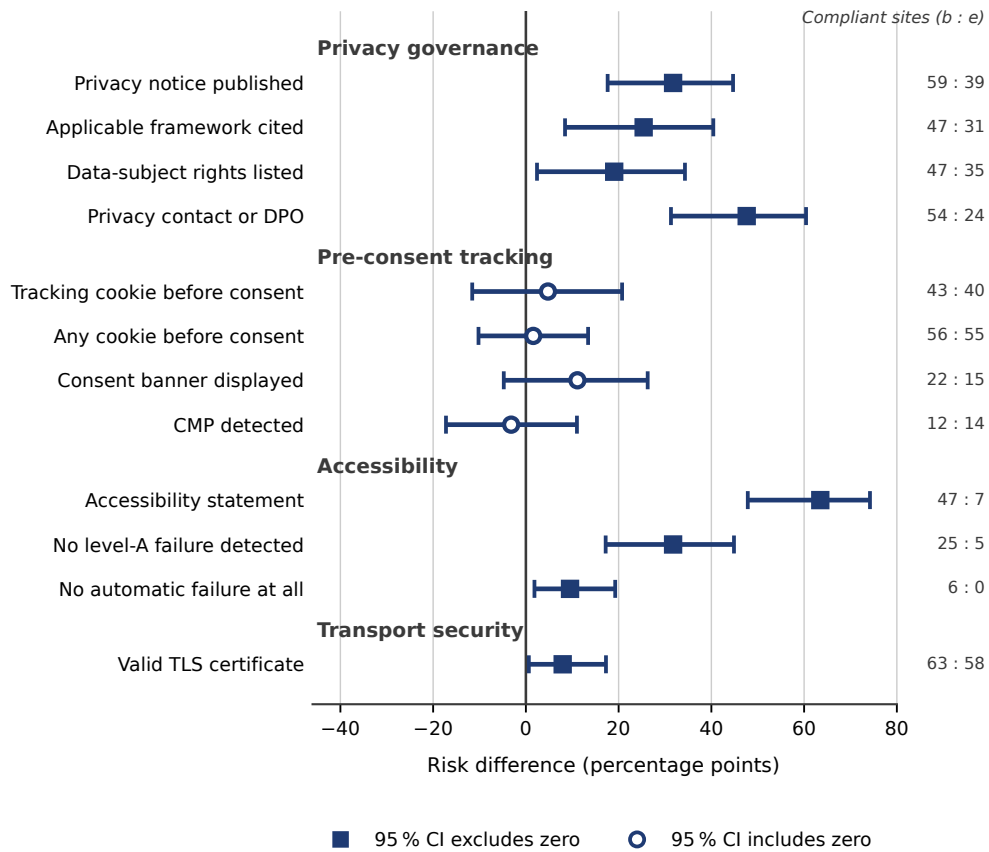


Fig. 1 Risk differences between the benchmark group and the Ecuadorian census, with 95% Newcombe intervals, grouped by domain. Filled squares mark intervals that exclude zero; open circles mark intervals that include it. Counts of compliant sites out of 63 are given at the right. Positive values indicate a higher proportion in the benchmark group.

Table 2 Benchmark group by region: institutions meeting each documentary indicator out of the regional total. Percentages and intervals are omitted deliberately: with n between 3 and 24 they would suggest a precision the data do not have.

Region	n	Notice	Framework	Contact/DPO	Accessibility
United States	24	24	17	23	22
United Kingdom	7	7	4	5	7
Continental Europe	10	10	9	10	8
China	6	2	2	3	0
Hong Kong SAR	3	3	3	2	3
Rest of Asia	5	5	4	4	0
Australia	5	5	5	4	4
Canada	3	3	3	3	3
Total	63	59	47	54	47

of benchmark sites and 87.3% of Ecuadorian sites; tracking cookies specifically were set at 68.3% and 63.5%. The difference on the tracking indicator was 4.8 percentage points in favour of Ecuador, with a 95% interval of $[-11.6, +20.7]$ percentage points and an adjusted p of 1.000. A visible consent banner was, in both groups, a minority phenomenon (34.9% and 23.8%), and a consent-management platform was detected slightly more often in Ecuador (22.2%) than in the benchmark group (19.0%), a difference well within sampling variation.

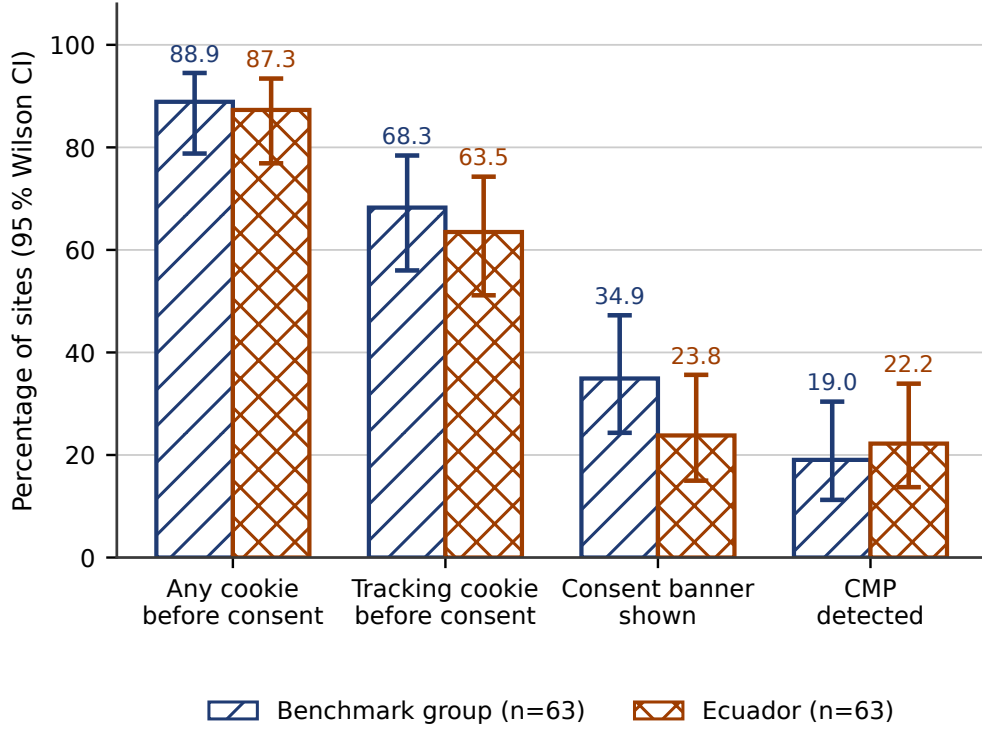


Fig. 2 Cookie state recorded before any interaction with a consent interface, with 95% Wilson intervals. All measurements were taken from a single vantage point in Ecuador in August 2026; a European visitor would not necessarily observe the same behaviour on the European sites of the benchmark group.

The absence of a detectable difference must not be read as a demonstration of equivalence. With 63 sites per group and proportions near 0.65, the smallest difference this design could detect with 80% power at $\alpha = 0.05$ is approximately 24 percentage points; detecting a true difference of 5 points would require roughly 1400 sites per group. Two one-sided tests confirm the point directly: equivalence is rejected at a margin of ± 10 percentage points ($p = 0.267$) and at ± 15 ($p = 0.113$), and is only supported at a margin of ± 20 ($p = 0.036$), which is too wide to be substantively interesting. Demonstrating equivalence within ± 10 points would require approximately

280 sites per group. The correct statement of this result is therefore that *the study found no evidence of a difference in pre-consent tracking, and had no power to establish that the groups are equivalent*.

A further and more important qualification concerns what a null result here can mean at all. Because all observations were taken from Ecuador, the benchmark figure conflates two possibilities that carry opposite legal readings: that European institutions do not condition their consent interfaces on the visitor’s location and simply track everyone, or that they do condition them and therefore show a European visitor a different interface from the one recorded here. The present design cannot distinguish the two, and Section 5 sets out why the distinction matters and how it should be resolved.

4.3 Declared accessibility and transport security (RQ3, part one)

An accessibility statement or a set of accessibility tools was published by 47 of the 63 benchmark institutions (74.6%) and by 7 of the 63 Ecuadorian institutions (11.1%), the largest single difference in the study (odds ratio 23.50, adjusted $p < 0.001$). Concrete statutory duties plausibly account for part of this gap: 22 of the 24 United States institutions publish a statement, in a jurisdiction with the ADA and Section 508 [59, 60], and 8 of the 10 continental European institutions do so under the European Accessibility Act [20]. Ecuador’s low figure is consistent with the regional lag documented over the past decade [1, 8].

Transport security was near-universal in both groups: a valid TLS certificate was present at all 63 benchmark sites and at 58 of the 63 Ecuadorian sites (92.1%; adjusted $p = 0.288$). We stress that this indicator measures confidentiality in transit only. A site with an impeccable certificate chain may transmit visitor data to any number of third parties, and five of the twelve indicators in Table 1 show precisely that pattern.

The LOTAIP transparency section required of Ecuadorian public bodies by the national transparency legislation was present at 47 of the 63 Ecuadorian institutions (74.6%). It is reported here for completeness because it is recorded in Appendix D; it is not used in any comparison, since it has no counterpart in the benchmark group.

4.4 Measured WCAG conformance (RQ3, part two)

Table 3 reports the audit results and Figure 3 the distribution of failing nodes by conformance level.

Neither group performed well. No Ecuadorian site and only 9.5% of benchmark sites produced no automatic failure at any enabled level, and the median number of distinct rules violated was four in Ecuador against two in the benchmark group. The distinction that matters, however, is where the failures fall. In the benchmark group 19% of failing nodes were at level A and 60% at level AAA, the latter dominated by enhanced contrast; in Ecuador 45% were at level A. Excluding the AAA rules entirely and re-normalising over levels A and AA preserves the ordering, 46.3% against 60.8% at level A, which indicates that the qualitative finding is not an artefact of having enabled a single demanding rule.

Table 3 Automatically detectable WCAG results (axe-core 4.13, August 2026, home page only). Percentages are of sites unless stated otherwise. The rule set is specified in Section 3.5.

Measure	Benchmark	Ecuador
Median distinct rules violated per site	2	4
Sites with no automatic failure at any level	9.5%	0.0%
Sites with no automatic level-A failure	39.7%	7.9%
Links without an accessible name (2.4.4, level A)	23.8%	81.0%
Insufficient contrast (1.4.3, level AA)	25.4%	79.4%
Images without a text alternative (1.1.1, level A)	15.9%	30.2%
<i>Distribution of failing nodes by level, all rules enabled</i>		
Level A	19%	45%
Level AA	22%	29%
Level AAA	60%	26%
<i>Same distribution with AAA rules excluded and re-normalised</i>		
Level A	46.3%	60.8%
Level AA	53.7%	39.2%

Note: the median is of distinct rules violated, not of failing nodes. Level percentages are of each group’s total failing nodes and are rounded to the nearest integer, which is why the benchmark distribution sums to 101%.

Two statements that appeared in an earlier version of this work are corrected here, because the data do not support them. It is *not* the case that no institution is free of automatic failures: six benchmark institutions were. And it is *not* the case that the benchmark group fails only at the level of excellence: 60.3% of benchmark sites produced at least one level-A failure, so the majority of the reference group does not clear the basic tier either. What the data support is narrower and still worth reporting: the two groups differ in the *distribution* of their failures, not in the presence of failures.

The single clearest instance is success criterion 2.4.4, which requires that every link have text a screen reader can announce; 81.0% of Ecuadorian sites and 23.8% of benchmark sites violated it. Insufficient contrast (1.4.3) followed the same pattern, 79.4% against 25.4%. The most affected principle in both groups is perceivability, through contrast; Ecuador adds an operability deficit through links and controls that is marginal in the benchmark group. These are the same dominant error types reported for other public-sector portals [3, 28].

Two cautions apply to the level-based measures. First, the distribution in Figure 3(a) is computed over failing *nodes*, which are not independent within a site: a single template with several thousand low-contrast elements can dominate its group’s distribution. Second, the distribution is compositional, so a group with more level-A failures must, arithmetically, show a smaller AAA share. Both are reasons to treat the level distribution as descriptive; the site-level measures in the upper half of Table 3 are the inferentially safe ones, and they point in the same direction.

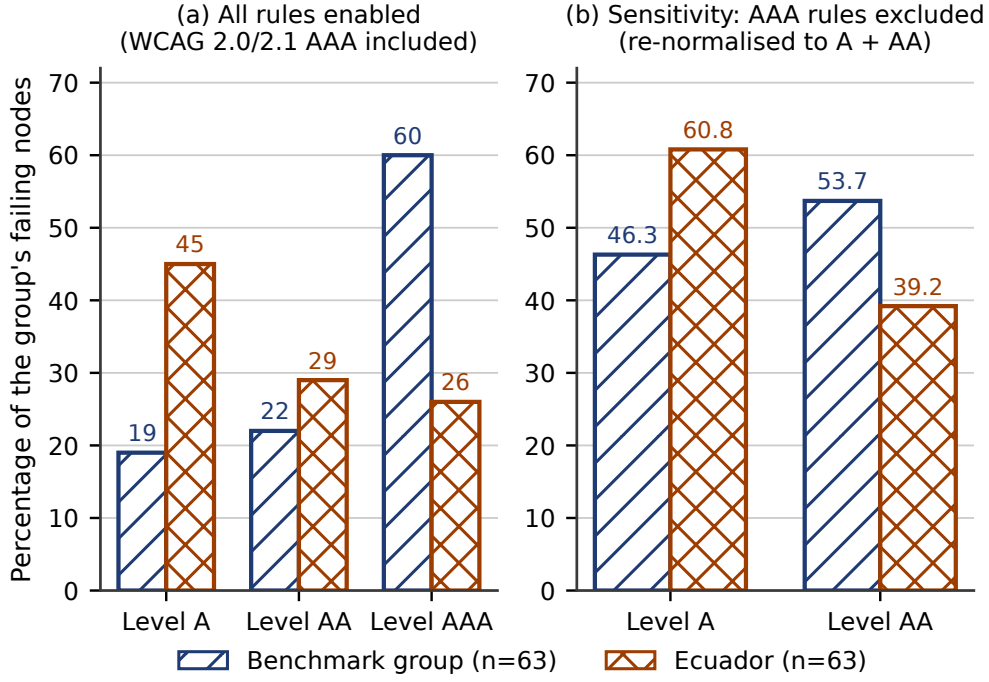


Fig. 3 Distribution of failing nodes by WCAG conformance level. Panel (a) uses the full rule set of this study, which includes the level-AAA rules of WCAG 2.0 and 2.1. Panel (b) excludes those rules and re-normalises over levels A and AA, showing that the ordering of the two groups at level A is preserved when the enhanced-contrast rule is not counted.

4.5 Do the two commitments co-occur? (RQ4)

Across all 126 sites, the Spearman correlation between the volume of accessibility failures and the number of tracking cookies is 0.04. That coefficient has a 95% confidence interval of $[-0.14, +0.21]$ and $p = 0.657$, and the design could only have detected a correlation of about 0.25 with 80% power. The pooled coefficient is therefore compatible with no association and with a weak positive association alike, and it cannot support a claim of independence.

The pooled coefficient is also the wrong estimand, because it mixes two populations whose marginal distributions differ on both variables. Table 4 gives the within-group cross-classification of the two dichotomised measures. Within the benchmark group the association is, if anything, slightly negative (odds ratio 0.54, Fisher $p = 0.408$); within the Ecuadorian group it is essentially absent (odds ratio 1.17, Fisher $p = 1.000$); the Mantel-Haenszel estimate adjusted for group is 0.66. None of these is distinguishable from no association at this sample size.

The defensible conclusion from RQ4 is a negative one, and it is worth stating precisely because the temptation is to overstate it. These data provide no evidence that institutions which invest in accessibility also restrain tracking, and no evidence of the reverse. They are not sufficient to establish that the two are independent; establishing

Table 4 Cross-classification of automatically detectable level-A conformance against pre-consent tracking, within each group. OR: odds ratio of conformance given restraint in tracking, with 95% interval.

Group	A^+T^-	A^+T^+	A^-T^-	A^-T^+	OR [95% CI]	p
Benchmark	6	19	14	24	0.54 [0.17, 1.68]	0.408
Ecuador	2	3	21	37	1.17 [0.18, 7.60]	1.000

Note: A^+ no automatically detectable level-A failure, A^- at least one; T^- no tracking cookie before consent, T^+ at least one. p from the two-sided Fisher exact test. In Ecuador, 2 institutions of 63 met both criteria. Under independence, $63 \times 0.079 \times 0.365 \approx 1.8$ would be expected, so this figure is exactly what chance predicts and carries no information about coupling.

that would require either a substantially larger sample or a continuous, better-powered pair of measures.

4.6 Whose location determines protection? (RQ5)

The four-vantage replication yielded 121 of the 126 sites measured successfully in all three passes at all four vantage points. The three control quantities behaved as required: the mean number of WCAG violations per site was 3.53, 3.51, 3.54 and 3.50 at the Ecuadorian, German, British and United States vantage points respectively, a spread of 1.1%, confirming that the pages themselves rendered identically. Between-pass stability of the tracking indicator ranged from 94.4% to 100%, so three passes were sufficient.

Pre-consent tracking, by contrast, differed by visitor location. It was observed at 67.8% of sites from Ecuador (95% CI 59.0–75.4), 65.3% from the United States, 62.8% from the United Kingdom and 60.3% from Germany (51.4–68.6). Cochran’s Q over the four vantage points gives $Q = 15.88$, 3 df, $p = 0.0012$. The paired comparisons in Table 5 locate the difference: Ecuador differs from Germany (9 discordant pairs, all in the same direction, $p = 0.0039$) and from the United Kingdom ($p = 0.031$), whereas Ecuador and the United States are indistinguishable ($p = 0.25$), as are the United Kingdom and Germany ($p = 0.25$).

Table 5 Exact McNemar tests for pre-consent tracking between vantage points ($n = 121$ sites observed at all four). b : sites tracking at the first vantage point but not the second; c : the converse. Only discordant pairs carry information.

Comparison	a	b	c	Discordant	p
Ecuador vs Germany	74	9	0	9	0.0039
Ecuador vs United Kingdom	77	6	0	6	0.031
United States vs Germany	73	7	1	8	0.070
United States vs United Kingdom	76	4	1	5	0.375
United Kingdom vs Germany	74	3	0	3	0.250
Ecuador vs United States	80	3	0	3	0.250

The mechanism is visible once the cookies are attributed to the vendor that sets them. Figure 4 shows the share of sites at which each vendor set at least one cookie before any consent interaction. Three vendors withheld cookies completely from European and British visitors while setting them for Ecuadorian and United States visitors on the same pages: Microsoft, whose Clarity, advertising and UET cookies appeared at 14 sites from Ecuador and 14 from the United States but at none from Germany or the United Kingdom; TapAd, 4 and 4 against zero; and StackAdapt, 2 and 2 against zero. The LinkedIn Insight Tag followed the same pattern in attenuated form, 14 and 14 against 3 and 7. Google Analytics, Meta and Adobe reduced their footprint only moderately and did so similarly at all four vantage points.

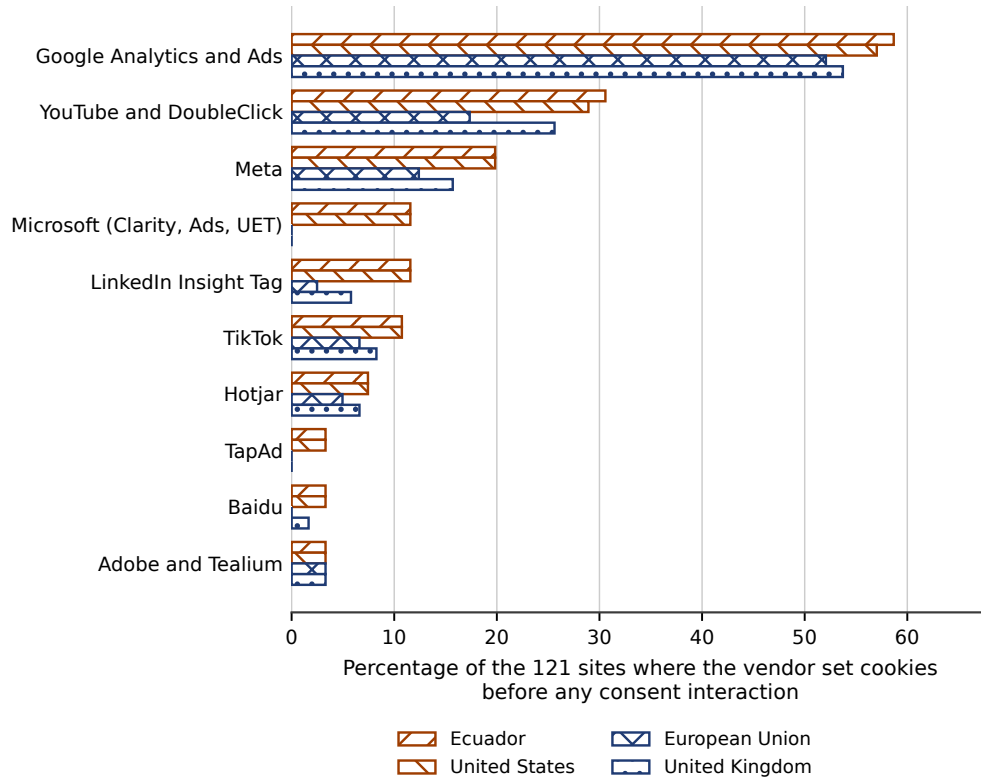


Fig. 4 Share of sites at which each advertising vendor set at least one cookie before any consent interaction, by vantage point ($n = 121$ sites, majority of three passes, August 2026). Ochre marks the two vantage points outside the European Economic Area and navy the two inside the perimeter that vendors apply. Vendors are ordered by their prevalence as seen from Ecuador; those present at fewer than four sites are omitted.

Two features of this pattern deserve emphasis because they are not what the institutional framing would predict. First, the suppression is not confined to European institutions: Ecuadorian, Chinese and United States universities that embed these

tags also stop setting the corresponding cookies when the visitor is in Germany or the United Kingdom. The behaviour therefore belongs to the vendor, not to the institution or to the law that binds it. Second, the United Kingdom is treated by most vendors as inside the protected perimeter although it is no longer in the European Economic Area, which is consistent with those vendors operating a country list rather than a legal test. TikTok and Baidu are the exceptions: both set materially more cookies for British visitors than for German ones, 10 against 8 sites and 2 against 0 respectively, so the perimeter is vendor-specific rather than common.

One caveat applies. Cookies not attributable to any advertising vendor also fell at the European and British vantage points, by 25.8% and 21.4% against Ecuador, where the United States vantage point differed by only 0.8%. That residual is smaller than the 43.6% and 37.4% fall in vendor cookies and it does not affect the rendering controls, but part of it is likely to be the Cloudflare bot-management cookie and the cookies set by consent platforms themselves, whose behaviour also varies with the visitor’s region. The residual is reported here rather than adjusted away.

5 Discussion

5.1 Interpretation

Three findings survive the analysis of Section 4, and they are of different kinds.

The first is a governance gap that is large, robust to multiple-comparison correction and consistent across four indicators. The Ecuadorian shortfall is not principally the absence of documents but the absence of the elements that make a document actionable: a named instrument, an enumeration of rights and, above all, an identified route through which a person can exercise them. The 47.6-point gap on the privacy contact is the single most consequential number in the study, because a right without a channel is not exercisable. That benchmark institutions perform well here is not evidence of superior ethics; it is the expected consequence of legal regimes that impose the duty explicitly, and it suggests that the same is achievable in Ecuador as the LOPDP and its regulation move into their enforcement phase [4, 50].

The second is an accessibility gap whose interesting property is its shape. Both groups fail; the majority of the benchmark group fails at the basic tier too. What differs is that Ecuadorian failures concentrate at level A, where the remedies are inexpensive and well understood, principally accessible link text and adequate contrast, and where the consequences for users of assistive technology are most severe. This is the most immediately actionable finding of the study and the one that requires no new legislation to act on.

The third is a null result on pre-consent tracking that must be reported as such. Observed from Ecuador, roughly two thirds of the institutions in both groups set tracking cookies before any consent interaction. The study found no difference; it also had no power to establish equivalence, and Section 4.2 states the bound explicitly. The finding that survives is therefore not “there is no gap” but “the gap, if any, is not large, and it is certainly not in the direction that the reputational ordering of the two groups would predict”.

5.2 Territorial scope, and who actually enforces it

An earlier version of this work explained the benchmark group’s behaviour by saying that the protection of the GDPR stops at the Union’s border, so that a student consulting a European university’s portal from Ecuador is as exposed as on a domestic site. That explanation was incorrect in law, and the four-vantage replication shows that it was also incorrect in fact, though not in the way we first supposed.

The legal point first. Article 3(1) of the GDPR applies to processing in the context of the activities of an establishment of a controller in the Union *regardless of whether the processing takes place in the Union or not*, and the European Data Protection Board has confirmed that the location of the data subject is irrelevant to that limb [16, 19]. A German or French university that sets marketing cookies in the browser of a visitor located in Ecuador is therefore acting within the material and territorial scope of the Regulation. Article 3(2) extends the Regulation to controllers not established in the Union that offer goods or services to data subjects *who are in* the Union, so that limb keys on the data subject’s location rather than the controller’s. The narrower and defensible point concerns a different instrument: Article 5(3) of the ePrivacy Directive, as transposed nationally, is articulated around the terminal equipment of a user, and the EDPB’s guidance on its technical scope is the appropriate reference [17, 18].

The measurement then shows something the institutional framing does not anticipate. Protection does vary with the visitor’s location, and substantially, but it is not the universities that vary their behaviour. It is the advertising vendors embedded in their pages. Microsoft, TapAd and StackAdapt set no cookies at all for a visitor in Germany or the United Kingdom and set them normally for a visitor in Ecuador or the United States, and they do so on Ecuadorian, Chinese and United States university sites as readily as on European ones. The institution is a bystander to this: it embeds a tag, and the tag decides.

Three consequences follow, and together they are the most original contribution this study can make.

The first is that compliance is being executed at the wrong layer. A duty that European law places on the controller, the university, is in practice discharged by a processor that the university did not configure and does not audit, on the basis of a geolocation lookup. That may satisfy the letter of the obligation for European visitors while leaving the controller without any knowledge of what its own site does.

The second is that the boundary is commercial, not legal. The United Kingdom left the European Economic Area but most vendors still place it inside the protected perimeter, while TikTok and Baidu do not. A person’s protection therefore depends on which country lists a set of private firms happen to maintain, and those lists are neither published, nor harmonised, nor subject to the review that a legal instrument would receive.

The third is distributive, and it is the point that connects this finding to the concern of universal access with which this article began. The population that receives no protection is the one located outside the vendors’ perimeter, which includes the entire international applicant pool of Latin America, Africa and much of Asia. An Ecuadorian applicant browsing a German university’s admission pages is tracked; a German applicant browsing an Ecuadorian university’s pages is not. The asymmetry

runs in the direction that compounds existing inequality, and it is invisible to any monitoring exercise conducted from inside the perimeter, which is where regulators, auditors and most researchers sit.

5.3 Implications for practice

For Ecuadorian institutions, the ordering implied by these results is unusually clear. The first action is to identify a privacy contact by role and address on the first-level notice; it is the cheapest intervention in the list and it closes the largest gap. The second is to name the applicable instrument and enumerate the rights the LOPDP confers, together with the procedure to exercise them, replacing generic references to “applicable legislation”. The third is to remediate the level-A accessibility failures, beginning with link text and contrast, which together account for the majority of the level-A nodes observed. The fourth, and the least urgent on the evidence collected here, is the consent interface: adding a banner without configuring it to block before consent changes the measured indicator without changing the user’s exposure, a substitution that the measurement literature has documented repeatedly [6, 53] and that the present data cannot rule out in either group.

For institutions in the benchmark group, the finding that 60.3% of them produce at least one level-A failure, and that two thirds set tracking cookies before consent when observed from outside the Union, is the relevant one. Reputational standing is not evidence of either accessibility or privacy performance. A further implication follows from Section 4.6 and applies to every institution in both groups: an administrator who tests their own site from their own office sees only what their own location entitles them to see. Verifying what an international applicant experiences requires testing from that applicant’s region, which is neither difficult nor expensive and is, on this evidence, not being done.

5.4 Implications for policy

Two implications follow for regulators. First, for the Ecuadorian data protection authority, the indicators that separate the two groups most sharply are exactly those that are verifiable at low cost from outside the institution, so a periodic supervised self-assessment against a published matrix is administratively feasible; the codebook in Appendix A is offered for that purpose. The accompanying risk is well known and should be anticipated: an indicator that becomes a target is satisfied at the surface, and the cookie banner is the clearest instance of an indicator that can be met without any change in behaviour.

Second, the pattern documented in Section 4.6 concerns European supervisory authorities rather than Ecuadorian ones, and it raises a question that supervision as currently practised does not ask. Where a controller’s compliance with Article 5(3) is delivered de facto by a processor’s geolocation logic, the controller cannot demonstrate accountability under Article 5(2) for conduct it neither configured nor observes. Supervisory testing conducted from inside the perimeter will find that controller compliant. We suggest that audits of consent behaviour be performed from outside the

European Economic Area as a matter of routine, and that the country lists operated by major tag vendors be treated as a supervisable object in their own right.

5.5 Threats to validity

Construct validity. Pre-consent tracking is measured uniformly across eleven jurisdictions whose obligations differ radically; in the United States there is no general prior-consent duty, and Ecuadorian law does not incorporate an equivalent of Article 5(3) of the ePrivacy Directive. Appendix B maps each indicator to the concrete obligation, if any, in each jurisdiction, and the results in Section 4.2 are stated as observed conduct, not as compliance findings. “No level-A failure detected” is a statement about the machine-testable subset and not a conformance claim [23]. Transport security is reported separately from data protection for the reason given in Section 4.3. Cookie-based measurement also misses `localStorage`, `IndexedDB`, fingerprinting, cookieless pixels and server-side tagging, so the tracking figures reported here are lower bounds.

Internal validity. The benchmark group is an elite stratum, not a sample of the world, so resource and jurisdiction effects are fully confounded; no third stratum was measured. Confounders that could be recorded were not, in particular budget, enrolment, site language, content management system and hosting provider, and the last of these is a plausible source of non-independence within the Ecuadorian group, where shared templates are common. The institution type recorded in Appendix D, public, co-financed or self-financed, is not analysed here.

Reliability. Documentary coding was performed without double coding and without an inter-coder reliability coefficient. The verification sub-study of Section 3.4 found two errors in 24 re-examined cells; this is a bound on nothing, but it is not reassuring, and the coding of items requiring judgement across several languages, among them Chinese, Japanese, Korean, German, French and Swedish, is the most exposed part of the design. Partial codes were treated as non-compliant, a convention that penalises institutions with segmented notices; the percentages were not recomputed under the alternative convention. Codes for non-verifiable sites remained in the denominator, which depresses the Ecuadorian percentages specifically, since all such cases occur in that group.

Vantage-point artefacts. The three non-Ecuadorian vantage points were reached through a commercial VPN and their egress addresses belong to data centres, so vendors that treat such addresses differently could in principle mimic a location effect. Three observations bound this concern: the rendering controls were identical across vantage points to within 1.1%; the German and United States endpoints share the same network type, so their contrast is free of the artefact; and one German endpoint was replaced by a second provider on a different autonomous system with an indistinguishable result. The residual fall in non-vendor cookies at the European and British vantage points, reported in Section 4.6, is not fully explained and is the weakest point of this part of the design. All four vantage points were also measured on a single day, so the replication describes vendor behaviour on that date.

Measurement stability. The documentary review remains a single observation. Pre-consent cookie state is not deterministic: it depends on A/B tests, ad auctions, CDN node and deployment timing. For the live audit this was addressed by the three passes

per vantage point, whose between-pass agreement ranged from 94.4% to 100%; for the group comparison of Section 4.2, which rests on the August single-pass measurement, it was not. Only the home page was audited, whereas WCAG conformance is defined over pages and complete processes, and the transactional paths this article invokes as its motivation, admission and enrolment, were not measured at all. The banner and CMP detectors are heuristics whose false-negative rates are unknown.

Statistical conclusion validity. Twelve comparisons were corrected by the Holm procedure and only the adjusted values are used to support claims; the regional breakdown is reported without inference for the reason given in Section 4.1. The equivalence margins in the TOST analysis were not pre-specified. The node-level accessibility distribution is subject to pseudo-replication and compositional constraint, as stated in Section 4.4. Failure counts were not normalised by page complexity, so a large portal will accumulate more failing nodes than a small one without necessarily being less accessible.

External validity. All observations were taken on one date from one country, and the results describe the state of 126 home pages in August 2026. Web content changes; no claim in this article should be read as a durable property of any institution.

6 Conclusions and future work

Measured against a benchmark of 63 top-ranked universities, the 63 Ecuadorian HEIs showed their widest and most robust gaps in privacy governance, above all in the identification of a privacy contact (38.1% against 85.7%) and in the publication of a first-level privacy notice (61.9% against 93.7%). In accessibility, both groups failed, but the failures were distributed differently: 45% of Ecuadorian failing nodes were at the basic level A against 19% in the benchmark group, an ordering preserved when the level-AAA rules are excluded. In pre-consent tracking, observed from Ecuador, the study found no difference between the groups and had no power to establish that they are equivalent. Within each group, measured accessibility and observed restraint in tracking were not associated.

Replicating the live audit from four vantage points then established that pre-consent exposure is governed by the visitor’s location rather than the institution’s. The same pages tracked 67.8% of the time for an Ecuadorian visitor and 60.3% for a German one (Cochran Q , $p = 0.0012$), and the suppression was executed by advertising vendors: Microsoft, TapAd and StackAdapt set nothing at all for European and British visitors while tracking Ecuadorian and United States visitors on the same pages, including on Ecuadorian and Chinese university sites.

These findings support a practical conclusion, a legal one and a distributive one. The practical conclusion is that Ecuador’s distinctive shortfalls are in governance and in basic-level accessibility, and both are remediable without new legislation. The legal conclusion is that a duty European law places on the controller is in practice being discharged by processors on the basis of unpublished country lists, which satisfies the visitor in Frankfurt while leaving the controller unable to demonstrate accountability for what its own site does. The distributive conclusion is the one that returns this article to its starting point: the applicants who receive no protection are those outside

the perimeter those lists describe, which is to say most of the world’s international applicant pool, and their exposure is invisible to anyone testing from inside it.

Four further items follow from the threats stated in Section 5.5: independent double coding of at least 30% of the sample with a Cohen’s κ per indicator and full reconciliation of discrepancies; extension of the audit to at least five pages per site, including an admission or enrolment path, with the page as the unit and the site as a random effect; more robust tracking measures than the cookie, in particular the count of distinct third-party domains contacted before consent; and an internal analysis of the Ecuadorian group by institution type, size and platform, for which the necessary variable is already recorded in Appendix D and which is likely to be of greater local value than any international comparison. To these the present results add a fifth: mapping the country lists operated by the major tag vendors, by measuring from one vantage point per candidate jurisdiction, so that the perimeter within which protection is delivered can be described empirically rather than assumed to coincide with any legal boundary.

Supplementary information. The audit script, its configuration, the raw per-site JSON output of the twelve multi-vantage passes and the location-verification records, the complete cookie inventory with its classification, the documentary matrix, the codebook and the analysis notebooks are deposited in a public repository. [AUTORES: depositar en Zenodo u OSF antes del envío y sustituir por el DOI asignado.]

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Author contributions. [AUTORES: completar según la taxonomía CRediT, por ejemplo:] G.G.-U.: conceptualisation, methodology, software, formal analysis, writing — original draft. A.Z.P.: investigation, data curation, writing — review and editing. E.D.-M.: investigation, validation, writing — review and editing. All authors read and approved the final manuscript.

Data availability. The datasets generated and analysed during the current study are available in the repository cited under Supplementary information. [AUTORES: sustituir por el DOI del depósito.]

Code availability. The audit script and the analysis code are available in the same repository.

Ethics approval and consent to participate. Not applicable. The study analysed publicly accessible web pages and involved no human participants, no animals and no personal data.

Use of generative artificial intelligence. [AUTORES: declarar si se empleó alguna herramienta de IA generativa en la redacción o el análisis y con qué alcance, o indicar expresamente que no se empleó. UAIS exige esta declaración cuando corresponda.]

ORCID. [AUTORES: añadir los tres identificadores ORCID en el sistema de envío de Springer.]

Appendix A Codebook and operational definitions

Table A1 gives the operational definition, the decision rule and the treatment of ambiguous cases for each indicator. The tracking-cookie taxonomy follows.

Table A1: Operational definitions of the coded indicators.

Indicator	Coded present when	Coded absent or partial when
Privacy notice	A document describing the processing of personal data is reachable from the home page or one click below it	P if its stated scope covers only part of the institution’s processing; absent if unreachable from the home page
Framework cited	The notice names a specific instrument, for example the LOPDP, the GDPR or the PDPO	Absent for generic formulations such as “applicable law”; P for a named framework that does not apply to the institution
Rights listed	At least three of access, rectification, erasure, restriction, objection and portability are enumerated, with a route to exercise them	Absent for an unenumerated mention of “your rights”; P if enumerated without any route
Privacy contact or DPO	A named role, a dedicated address or a dedicated form for privacy matters is given	Absent for a general institutional contact form or switchboard number
Accessibility declared	An accessibility statement, an accessibility policy or an accessibility toolbar is published	Absent when accessibility is mentioned only inside another document
Transport security	The home page is served over HTTPS with a chain that a current browser accepts without warning	Absent for expired, self-signed or incomplete chains
Tracking before consent	At least one cookie whose name matches the taxonomy below is present before any interaction with a consent interface	Absent when no matching cookie is present
No level-A failure	axe-core reports no violation carrying a level-A tag	Absent when at least one level-A violation is reported

Tracking-cookie taxonomy. A cookie was classified as belonging to an advertising or analytics vendor when its name matched one of the following patterns, applied as

regular expressions to the cookie name. The list was extended after the multi-vantage replication of Section 3.6; the families added at that point are marked with an asterisk, and every figure in this article uses the extended list, applied by reclassifying the stored cookie names.

- *Google Analytics and Ads*: $\sim_ga$, $\sim_gid$, $\sim_utm$, $\sim_gcl$, $\sim_gac$; *Privacy Sandbox**: $\sim\text{receive-cookie-deprecation}$.
- *YouTube and DoubleClick*: $\sim YSC$, $\sim VISITOR_INFO$, $\sim VISITOR_PRIVACY$, $\sim_Secure-YNID$, $\sim_Secure-ROLLOUT$, $\sim IDE$, $\sim test_cookie$, $\sim DSID$.
- *Meta*: $\sim_fbp$, $\sim_fb$, $\sim_fr$. *TikTok*: $\sim_tt$, $\sim_ttp$, $\sim_ttcsid$.
- *Microsoft Clarity, Advertising and UET*: $\sim_clck$, $\sim_clsk$, $\sim MUID$, $\sim MR$, $\sim SM$, $\sim SRM_B$, $\sim ANONCHK$, $\sim CLID$, $\sim_uet$.
- *LinkedIn Insight Tag**: $\sim bcookie$, $\sim bscookie$, $\sim lidc$, $\sim li_sugr$, $\sim li_gc$, $\sim UserMatchHistory$, $\sim AnalyticsSyncHistory$.
- *TapAd**: $\sim TapAd$. *StackAdapt**: $\sim sa-user-id$. *Snapchat**: $\sim_scid$, $\sim_sc_at$.
- *X and Twitter*: $\sim muc_ads$, $\sim personalization_id$, $\sim_twpid$.
- *Hotjar*: $\sim_hj$. *Siteimprove*: $\sim nmstat$. *Matomo*: $\sim_pk$. *PixelYourSite and Sourcebuster*: $\sim pys$, $\sim sbjs$. *Pinterest*: $\sim_pin$.
- *Adobe and Tealium*: $\sim AMCV$, $\sim s$, $\sim utag$, $\sim mbox$; *ContentSquare**: $\sim_cs$.
- *Baidu*: $\sim Hm_lvt$, $\sim Hm_lpvt$, $\sim HMAccount$, $\sim BAIDUID$. *Quantcast*: $\sim_qca$. *LiveIntent*: $\sim_lc2$, $\sim ln_or$.

The list is name-based and therefore conservative: a vendor using an unlisted or randomised cookie name is not counted, which is one reason the figures in Section 4.2 are lower bounds. The consequence of the initial omissions is quantified in Section 4.6: under the original list the fall in vendor cookies at the European vantage point was 33.0%, and under the extended list it is 43.6%.

Appendix B Indicator-by-jurisdiction mapping

Table B1 states, for each jurisdiction in the study, the instrument accepted as the applicable framework and whether the jurisdiction imposes a general duty of prior consent for non-essential cookies. It is the basis for the distinction, maintained throughout Section 4, between observed conduct and legal non-compliance. Figure E2 in Appendix E places these regimes on a common timeline.

Table B1: Applicable framework and prior-consent duty by jurisdiction.

Jurisdiction	Instrument accepted as applicable framework	General prior-consent duty for non-essential cookies
Ecuador	LOPD and its implementing regulation [4, 50]	No equivalent of Art. 5(3) ePrivacy; consent is required for processing, not as a storage-access rule
European Union	GDPR [19] and national transpositions of the ePrivacy Directive [18]	Yes, under Art. 5(3) ePrivacy as transposed [17]
United Kingdom	UK GDPR and Data Protection Act 2018 [49]	Yes, under the national ePrivacy transposition

Continued on next page

(Table B1, continued)

Jurisdiction	Instrument accepted as applicable framework	General prior-consent duty for non-essential cookies
Switzerland	Federal Act on Data Protection [55]	No general prior-consent rule; transparency and opt-out apply
United States	CCPA [7] for California; FERPA [58] and HIPAA [61] govern education records and protected health information respectively and do not govern the public portal	No general prior-consent duty; opt-out model
Canada	Provincial public-sector statutes for public universities [32, 33, 42]; PIPEDA [47] for commercial activity	No general prior-consent duty; implied consent recognised
China	PIPL [44]	Consent required for non-essential processing
Japan	APPI [24]	No general prior-consent duty for cookies as such
South Korea	PIPA [43]	Consent required for non-essential processing
Singapore	PDPA [48]	Deemed consent model; no general prior-consent duty
Australia	Privacy Act 1988 [46]	No general prior-consent duty
Hong Kong SAR	PDPO [34]	No general prior-consent duty

The consequence for interpretation is direct: in the majority of the jurisdictions studied, setting a tracking cookie before consent is not in itself unlawful. The indicator is retained because it measures the exposure of the visitor, which is the concern of this article, but it is not, and is nowhere presented as, a compliance rate.

Appendix C Documentary matrix, benchmark group

Legend: ✓ present; – absent; **P** partial, meaning limited scope or a generic framework; *NV* not verifiable by this method. Rankings: QS 2026, THE 2026 and ARWU 2025, giving the position, with “–” for absence from that ranking’s top 75. Country in ISO 3166-1 alpha-2. Columns: Not. (first-level privacy notice), Frm. (specific legal instrument named), Rts. (data-subject rights enumerated), DPO (privacy contact or data protection officer identified), Ckp. (a dedicated cookie policy is published, a documentary indicator distinct from the live banner measurement of Section 4.2), Acc. (accessibility statement or tools declared). Coding date: August 2026. Cells marked † were corrected after the manual re-verification described in Section 3.4.

Table C1: Documentary matrix of the 63 benchmark institutions.

#	Abbr.	C	QS	THE	ARWU	Not.	Frm.	Rts.	DPO	Ckp.	Acc.
1	MIT	US	1	2	3	✓	✓	✓	✓	–	✓
2	Stanford	US	3	5	2	✓	✓	✓	✓	✓	✓
3	Harvard	US	5	5	1	✓	–†	–	–	–	✓
4	Oxford	GB	4	1	6	✓	✓	✓	✓	✓	✓

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(Table C1, continued)

#	Abbr.	C	QS	THE	ARWU	Not.	Frm.	Rts.	DPO	Ckp.	Acc.
5	Cambridge	GB	6	3	4	✓ [†]	P	—	—	✓	✓
6	Caltech	US	10	7	9	✓	✓	—	✓	—	✓
7	UC Berkeley	US	17	9	5	✓	✓	—	✓	—	✓
8	Princeton	US	25	3	7	✓	✓	✓	✓	—	✓
9	Imperial	GB	2	8	26	✓	✓	P	—	✓	✓
10	U Chicago	US	13	15	10	✓	P	✓	✓	—	—
11	ETH Zurich	CH	7	11	22	✓	✓	✓	✓	—	✓
12	Yale	US	21	10	11	✓	✓	P	✓	—	✓
13	UPenn	US	15	14	14	✓	P	✓	✓	—	✓
14	UCL	GB	9	22	14	✓	P	✓	✓	✓	✓
15	Cornell	US	16	18	12	✓	P	✓	✓	—	✓
16	Tsinghua	CN	17	12	18	✓	✓	✓	✓	—	—
17	Peking	CN	14	13	23	P	P	—	—	—	—
18	Johns Hopkins	US	24	16	19	✓	✓	✓	✓	—	✓
19	Columbia	US	38	20	8	✓	✓	✓	✓	✓	✓
20	Toronto	CA	29	21	25	✓	✓	✓	✓	—	✓
21	UCLA	US	46	18	16	✓	✓	✓	✓	—	✓
22	NUS	SG	8	17	56	✓	✓	✓	✓	—	—
23	U Tokyo	JP	36	26	31	✓	✓	✓	—	—	—
24	TUM	DE	22	27	45	✓	✓	✓	✓	—	—
25	Melbourne	AU	19	37	38	✓	✓	✓	✓	—	—
26	Edinburgh	GB	34	29	37	✓	✓	✓	✓	—	✓
27	EPFL	CH	22	35	44	✓	✓	✓	✓	—	✓
28	Michigan	US	45	23	33	✓	—	—	✓	—	—
29	Northwestern	US	42	30	31	✓	P	✓	✓	✓	✓
30	Fudan	CN	30	36	41	—	—	—	—	—	—
31	PSL	FR	28	48	34	✓	✓	✓	✓	—	—
32	HKU	HK	11	33	67	✓	✓	✓	✓	—	✓
33	Zhejiang	CN	49	39	24	✓	P	✓	✓	—	—
34	NYU	US	55	31	28	✓	✓	✓	✓	—	✓
35	SJTU	CN	47	40	30	—	—	—	—	—	—
36	KCL	GB	31	38	61	✓	P	✓	✓	✓	✓
37	UCSD	US	66	47	20	✓	✓	✓	✓	—	✓
38	LMU	DE	58	34	42	✓	✓	✓	✓	—	✓
39	Duke	US	62	28	46	✓	✓	✓	✓	—	✓
40	Manchester	GB	35	56	46	✓	✓	P	✓	—	✓
41	UBC	CA	40	45	53	✓	✓	P	✓	—	✓
42	Sydney	AU	25	53	72	✓	✓	✓	✓	—	✓
43	Paris-Saclay	FR	70	68	13	✓	✓	✓	✓	✓	✓
44	Kyoto	JP	57	61	46	✓	P	✓	✓	—	—
45	Illinois	US	70	41	53	✓	✓	✓	✓	—	✓
46	UT Austin	US	68	50	49	✓	✓	✓	✓	—	✓
47	UW	US	—	25	17	✓	✓	✓	✓	—	✓
48	NTU	SG	12	31	—	✓	✓	✓	✓	—	—
49	McGill	CA	27	41	—	✓	✓	✓	✓	✓	✓
50	CUHK	HK	32	41	—	✓	✓	✓	—	—	✓
51	CMU	US	52	24	—	✓	✓	✓	✓	—	✓
52	Wisconsin	US	—	53	36	✓	✓	—	✓	✓	✓
53	USTC	CN	—	51	40	P	✓	✓	✓	—	—
54	WUSTL	US	—	67	26	✓	—	—	✓	—	✓
55	Monash	AU	36	58	—	✓	✓	—	—	—	✓
56	SNU	KR	38	58	—	✓	✓	✓	✓	—	—
57	Heidelberg	DE	—	49	51	✓	✓	✓	✓	—	✓
58	HKUST	HK	44	58	—	✓	✓	✓	✓	—	✓
59	Karolinska	SE	—	53	50	✓	P	✓	✓	✓	✓
60	TU Delft	NL	47	57	—	✓	✓	✓	✓	✓	✓

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(Table C1, continued)

#	Abbr.	C	QS	THE	ARWU	Not.	Frm.	Rts.	DPO	Ckp.	Acc.
61	ANU	AU	32	73	–	✓	✓	P	✓	–	✓
62	KU Leuven	BE	60	46	–	✓	✓	✓	✓	✓	✓
63	Queensland	AU	42	–	65	✓	✓	✓	✓	–	✓

Appendix D Documentary matrix, Ecuadorian census

Legend: ✓ present; – absent; **P** partial, meaning limited scope, for example postgraduate programmes, admissions or a mobile application only; *NV* not verifiable by this method, meaning the site could not be retrieved, the document was not machine-readable, or the element is rendered dynamically and could not be detected. Type: P public, C co-financed, A self-financed. Columns: Notice, LOPDP (the law is named expressly), Rights, DPO, Acc. (accessibility statement or tools declared), Transp. (LOTAIP transparency section published). Institutions created in 2024 or later are marked ‡. Coding date: August 2026. The live cookie measurements are reported in aggregate in Section 4.2 and per site in the deposited dataset; they are not reproduced here, because the documentary and live indicators are different constructs and presenting them in one grid invited the confusion this table avoids.

Table D1: Documentary matrix of the 63 Ecuadorian HEIs.

#	Abbr.	Type	Notice	LOPDP	Rights	DPO	Acc.	Transp.
1	EPN	P	✓	✓	✓	–	–	✓
2	ESPOL	P	✓	✓	✓	✓	–	✓
3	ESPOCH	P	–	–	–	–	–	✓
4	ESPAM MFL	P	–	–	–	–	–	✓
5	ESPE	P	✓	✓	✓	✓	✓	✓
6	UCE	P	<i>NV</i>	–	–	–	–	✓
7	UCuenca	P	✓	✓	✓	–	–	✓
8	UG	P	✓	✓	✓	✓	–	✓
9	UNL	P	–	–	–	–	–	✓
10	UAE	P	–	–	–	–	–	✓
11	UTA	P	✓	<i>NV</i>	<i>NV</i>	<i>NV</i>	<i>NV</i>	✓
12	UTB	P	✓	✓	✓	✓	–	✓
13	UTC	P	✓	–	–	–	–	<i>NV</i>
14	UTMACH	P	✓	✓	✓	✓	–	✓
15	UTM	P	–	–	–	–	–	✓
16	UTN	P	P	✓	✓	–	–	✓
17	UTEQ	P	P	✓	–	–	–	✓
18	UTELVT	P	–	–	–	–	–	✓
19	UEB	P	✓	<i>NV</i>	<i>NV</i>	<i>NV</i>	✓	✓
20	UNESUM	P	P	<i>NV</i>	–	–	–	✓
21	UNEMI	P	✓	✓	✓	–	–	✓
22	UEA	P	✓	–	✓	–	–	✓
23	UPSE	P	P	✓	✓	✓	P	✓
24	ULEAM	P	–	–	–	–	–	✓
25	UNACH	P	✓	✓	✓	✓	✓	✓
26	UNAE	P	–	–	–	–	–	✓
27	UPEC	P	–	P	–	–	–	✓
28	Yachay Tech	P	–	–	–	–	–	✓
29	Ikiam	P	✓	✓	✓	✓	–	✓
30	UArtes	P	✓	✓	✓	✓	–	✓
31	Amawtay Wasi	P	–	–	–	–	–	✓

Continued on next page

(Table D1, continued)

#	Abbr.	Type	Notice	LOPDP	Rights	DPO	Acc.	Transp.
32	USECIPOL [‡]	P	P	—	—	—	—	✓
33	IAEN	P	✓	✓	✓	—	✓	✓
34	UASB	P	✓	✓	✓	✓	—	✓
35	FLACSO	P	—	—	—	—	—	✓
36	PUCE	C	✓	✓	✓	✓	—	✓
37	UCSG	C	✓	✓	✓	✓	—	✓
38	UCACUE	C	✓	✓	✓	✓	✓	✓
39	ULVR	C	—	—	—	—	—	✓
40	UTPL	C	✓	✓	✓	✓	—	NV
41	UTE	C	✓	—	—	—	✓	NV
42	UDA	C	—	—	—	—	—	✓
43	UPS	C	✓	✓	✓	—	—	✓
44	UISEK	A	✓	—	✓	—	—	—
45	UEES	A	✓	✓	✓	✓	—	—
46	USFQ	A	✓	✓	✓	✓	—	—
47	UDLA	A	✓	✓	✓	✓	—	—
48	UIDE	A	✓	✓	✓	✓	—	✓
49	UNIANDES	A	✓	—	✓	—	—	—
50	UPACÍFICO	A	NV	—	—	—	—	—
51	UTI	A	✓	—	✓	—	—	✓
52	Casa Grande	A	✓	✓	✓	✓	—	—
53	UTEG	A	✓	✓	✓	✓	—	✓
54	UISRAEL	A	✓	NV	NV	NV	NV	NV
55	UDET	A	—	—	—	—	—	—
56	UMET	A	✓	NV	NV	NV	—	✓
57	U. Otavalo	A	NV	NV	—	—	—	NV
58	USGP	A	✓	✓	✓	✓	—	✓
59	U. Hemisferios	A	✓	✓	✓	✓	—	✓
60	UNIB.E	A	✓	—	✓	—	—	—
61	ECOTEC	A	✓	✓	✓	✓	—	✓
62	UDR	A	—	—	—	—	—	—
63	UBE	A	✓	✓	✓	✓	✓	—

Appendix E Supplementary figures

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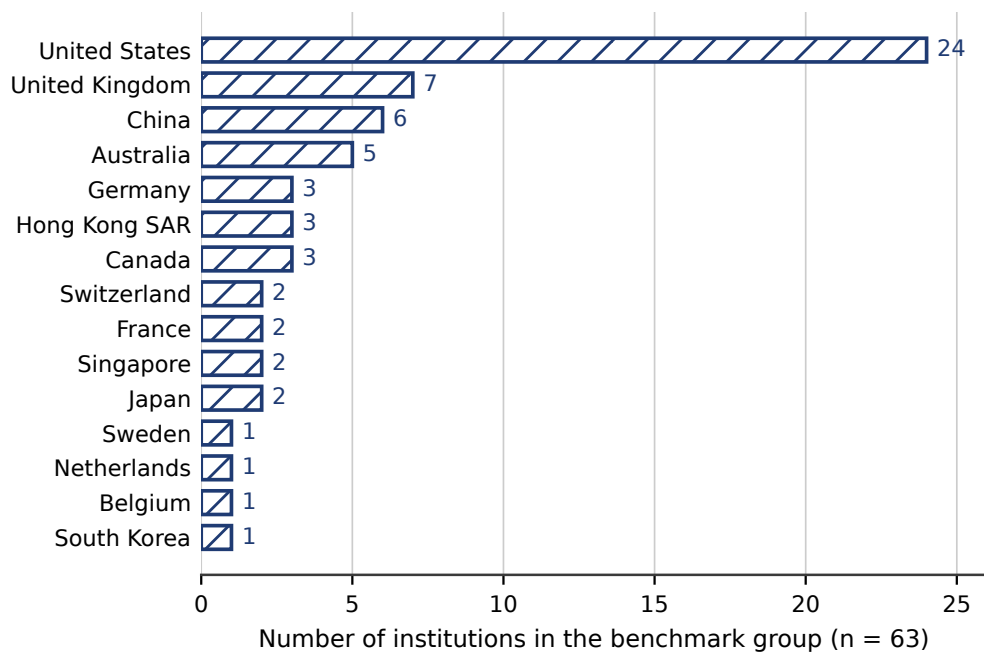


Fig. E1 Composition of the benchmark group by country. The group is the extreme upper tail of three international rankings and is not a sample of the world's universities; see Section 3.1.

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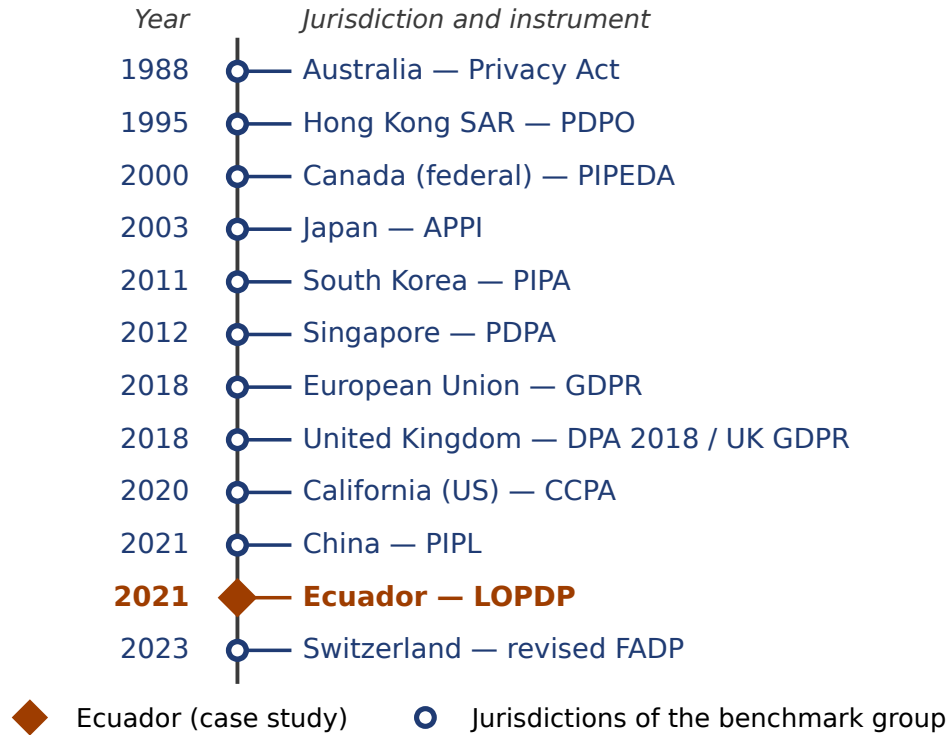


Fig. E2 Year in which the current general data-protection regime took effect in each jurisdiction of the study. For the United States, which has no comprehensive federal statute, California’s CCPA is used. Ecuador belongs to the most recent wave, alongside China.

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